higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# NATIONAL CERTIFICATE

# MECHANICAL DRAWING AND DESIGN N6

(8090086)

**28 November 2019 (X-Paper)**
**09:00–13:00**

**OPEN-BOOK EXAMINATION**

**This question paper consists of 6 pages.**

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# DEPARTMENT OF HIGHER EDUCATION AND TRAINING
# REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MECHANICAL DRAWING AND DESIGN N6
TIME: 4 HOURS
MARKS: 100

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**INSTRUCTIONS AND INFORMATION**

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. A correct answer is worth only ONE mark.
5. Show ALL the steps to achieve as many marks as possible. An incorrect answer which has to be carried forward will receive fair consideration.
6. Candidates may use their personal notes.
7. Write neatly and legibly.

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## QUESTION 1

FIGURE 1 shows a gear reduction unit. Pinion A has 22 teeth with a tooth module of 8 mm and delivers 150 kW at 1 200 r/min clockwise. The overall transmission ratio is approximately 8:1. The centre distances are nearly the same for each step. The reduction at each step is approximately equal. Assume that all teeth are full depth, 20° involute.

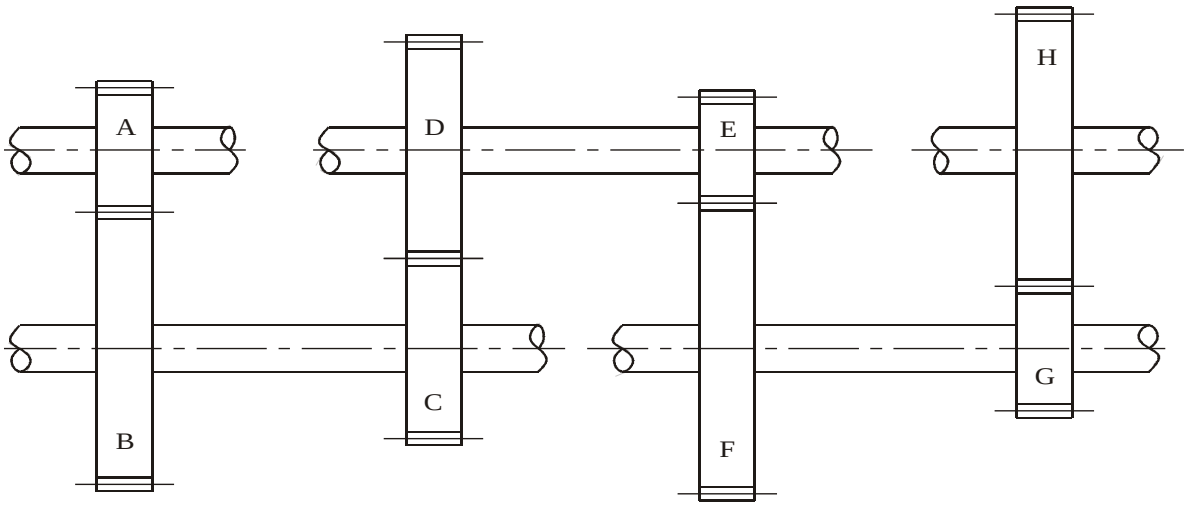



**FIGURE 1**

Calculate:

| | | |
|---|---|---|
| 1.1 | The approximate reduction at each step | (2) |
| 1.2 | The number of teeth on gear B | (1) |
| 1.3 | The centre distance of A and B | (1) |
| 1.4 | The number of teeth on gears C and D if the module is 4 mm | (3) |
| 1.5 | The number of teeth on gears E and F if the module is 6 mm | (2) |
| 1.6 | The number of teeth on gears G and H if the module is 5 mm | (3) |
| 1.7 | The torque on gear H | (5) |

**[17]**

## QUESTION 2

A part of a machine mechanism is shown in FIGURE 2. Using Rankine's theory, calculate the dimensions of the two struts A–C and B-C if a horizontal load of 60 kN is applied to point C. Strut A–C is 410 mm long and strut B–C is 480 mm in length. Assume hinged ends for both. The distance A–B is 400 mm and uses a factor of safety of 4.

If the material to be used has a yield stress of 320 MPa, calculate the material thickness required if the rods are of an I-section profile with the depth being 7,5 times the thickness and the width 2,5 times the thickness. Use a Rankine constant of 1/7 500.

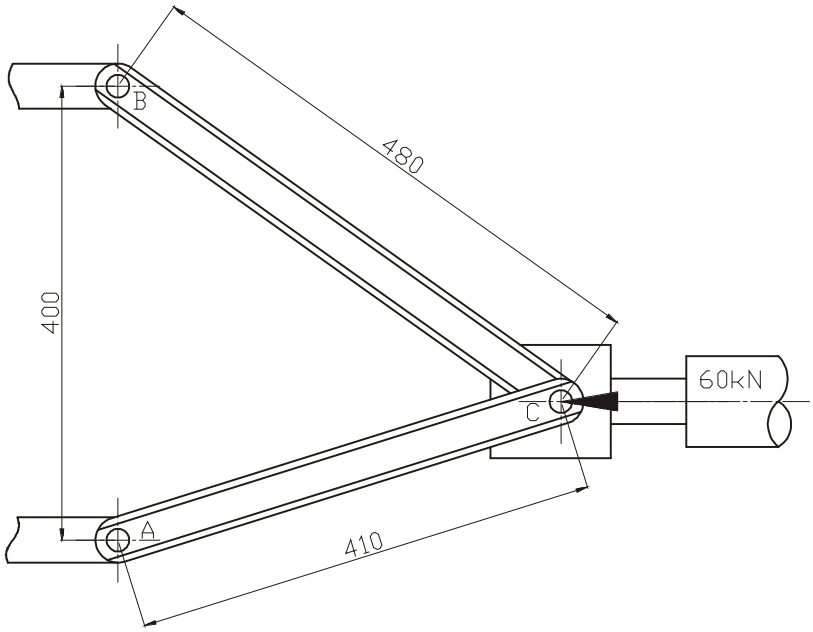



**FIGURE 2** **[25]**

## QUESTION 3

In FIGURE 3, driving pinion a has 30 teeth and must transmit 25 kW at 800 revolutions per minute to the driven gear c, with 40 teeth. Gear b is an idler gear and has 50 teeth. All the gears are 200 full-depth involute with a 6 mm module.

Calculate the resultant force on the idler gear. Use a basic stress of 80 MPa and determine the tooth width of gears a and b. Use the Lewis formula. Test the answer for acceptability.

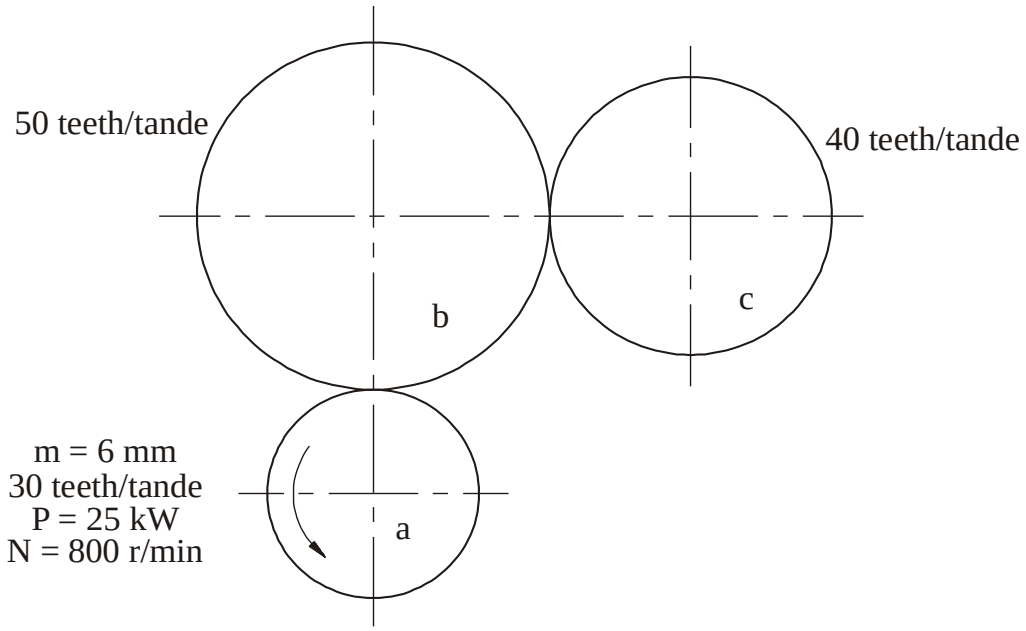



**FIGURE 3** **[20]**

## QUESTION 4

A shaft rotating at 300 r/min with the centrally keyed wheel of a worm drive, is shown in FIGURE 4. The shaft, which has a diameter of 45 mm, is supported in two journal bearings and a conical thrust bearing. The assembly is placed in a well-ventilated area where the ambient temperature is 25 °C and is lubricated with SAE oil. A force of 28 kN acting at an angle of 50° to the horizontal, is present on the wheel. The pressure on the journal bearing material cannot exceed 1,2 MPa and the temperature of the bearing surface is limited to 50 °C. Assume that the journal diameter to diametral clearance ratio (D/C) is 1 000.

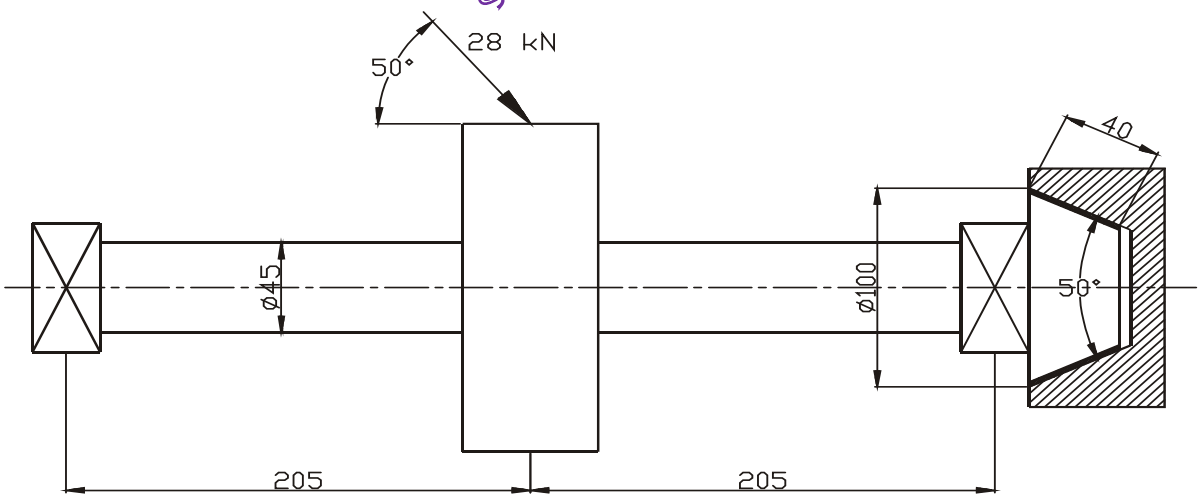



**FIGURE 4**

Calculate:

4.1 The radial load on the journal bearings (1)

4.2 The length of each journal (1)

4.3 The heat dissipated by each journal bearing (3)

4.4 The coefficient of friction (3)

4.5 The SAE number of the lubricating oil (4)

4.6 The side thrust (1)

4.7 The inside diameter of the conical thrust bearing to take this side thrust if it is made of the same material as the journals, but has an outer diameter of 100 mm and a width of 40 mm with a 50° cone angle (3)

4.8 The frictional torque that occurs at the thrust bearing using the uniform pressure theory (Use the coefficient of friction calculated in QUESTION 4.4.) (2)

4.9 The total power lost in the journals and the thrust bearing (2)

4.10 The quantity of lubricating oil needed in litres/min if its specific heat capacity is 3,2 kJ/kg °C and its density is 850 kg/m$^3$ (3)

**[23]**

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**QUESTION 5**

A solid shaft rotating at 450 r/min is seen in FIGURE 5. It is supported by two bearings 650 mm apart. 30 kW is supplied to the shaft through a 400 mm diameter pulley located 325 mm to the right of the left-hand bearing. The power is transmitted from the shaft through an 80 mm diameter pinion located 325 mm to the right of the right-hand bearing. The belt drive is at an angle of 60° below the horizontal and to the right. The ratio of belt tensions is 4:1 and the pulley has a mass of 6 kg. The gear wheel tooth form is 20° and it meshes with another gear directly above the shaft. The shaft material has ultimate shear strength of 400 MPa and a factor of safety of 4 is used.

Design a suitable shaft diameter.

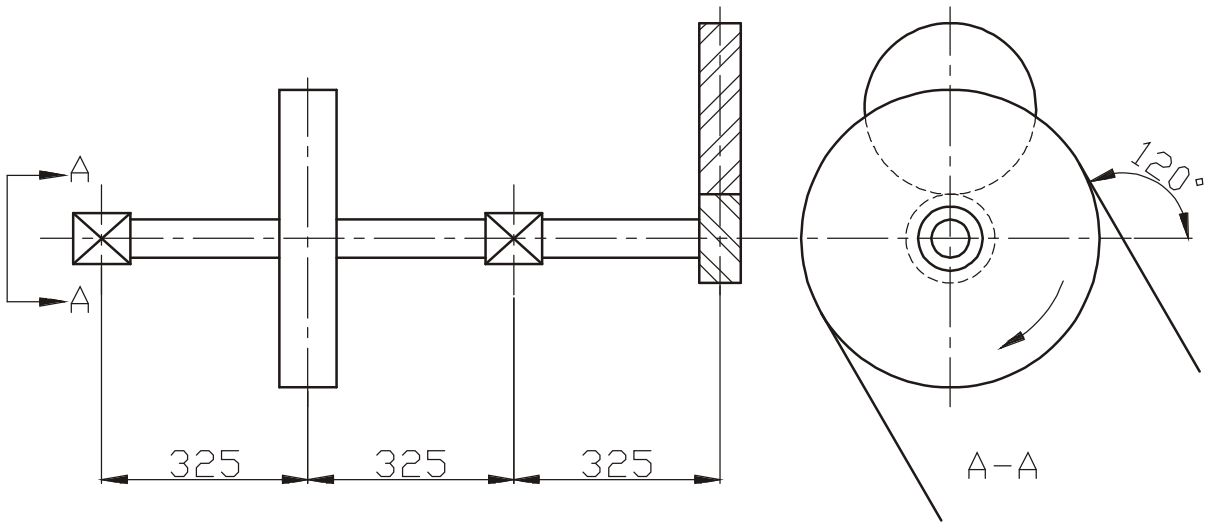



**FIGURE 5** **[15]**

**TOTAL: 100**

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